Khojki OCR | Technical retrospective

Printed-page prototype - measured limitations - 31 October 2026

LATEST FROZEN TEST CER	PRIOR FROZEN TEST CER	CPU PAGE TIME
21.14%	**14.41%**	**2.91 s**

Pipeline and measured outcome

Page/PDF -> background-relative ink and frame removal -> line peaks -> text/number chunks -> two 1.06M-parameter CNN-BiLSTM-CTC recognizers -> beam search -> UTF-8 and offline review. HarfBuzz/FreeType shape Noto synthetic examples; 15 real printed lines adapt the selected models. Inference needs no API or retraining.

Latest frozen test: 26/123 edits, 21.14% CER on five real lines; original baseline: 28/123 (22.76%). The preceding five-line test was 17/118 (14.41%), versus baseline 25/118 (21.19%). Development reached 14/405 (3.46%); the gap is real. Labels are non-expert, chart-transcribed and frozen before prediction inspection. Same-book/page relationships prevent collection-wide claims. Predeclared alternatives on the latest test give 20.16-25.20% CER, not a confidence interval.

Errors by stage

Latest test: **0 missed selected lines; 2 space-involving edits; 24 other recognition edits; 0 invalid code points / UTF-8 failures**. Automatic and checked crops give identical predictions. Space edits are layout/recognition symptoms, not a proven causal allocation. Repeated errors include inserted SA and vowel U (two each). The revised detector also separates a short refrain previously merged on page 20. Unlabelled detections are not scored.

Experiments that failed

Synthetic validation: 1.31% CER, but initial real-line CER: 48%; contrast/column splitting: 29.33%. On 405 development characters: 300-step control 6.91%, extended control 4.44%, 40% word recombination 8.64%, both 64-pixel pilots 8.64%, five extra labels 5.68%. Longer packed recurrence plus ensemble/filter reached 3.46%. A final targeted-data round gave 24/453 and 26/453 edits versus incumbent 20/453; both pilots were rejected. Histories include failed checkpoints, not just minima.

Encoding choices and validation

Vowel signs use U+1122C-U+11233; anusvara U+11234; independent AA U+11201. O/AU are atomic U+11232/U+11233. Word separators use U+1123A; printed digits use Gujarati U+0AE6-U+0AEF. ASCII space marks a distant number column. CER counts Unicode scalar values, including signs/digits. Inventory, orphan/adjacent signs, atomic vowels, NFC changes and UTF-8 round trips are checked. Structural filtering cannot certify a plausible wrong letter.

Highest-value improvement

Obtain **200-500 independently reviewed printed lines** spanning books, typefaces, damage and rare signs; double-annotate a subset, reserve entire books and measure a 50/100/200/500-line curve. This is a planning estimate. Further tuning against a tiny, uncertain development set is weak evidence. Historical TA/MA and NA/VA need adjudication. Some output classes have no positive examples. Initial 31-line authoring/freezing spanned about 14 minutes plus prior chart study and later review; this is not human-expert productivity.

Scale and handwriting

The selected model processes a 16-line page in **2.91 s on a two-thread Colab CPU**, plus 0.044 s loading; startup/PDF rendering/review are excluded. Approximately 137 s for 47 comparable printed pages, or 2.91N s for N pages. The mixed collection will not share this accuracy. T4 was about 1.15 units/hour; setup/idle time dominated. Exports were hash-verified; final balance 167.31 units, zero active sessions. Handwriting needs baseline/polygon detection, writer-disjoint evaluation and roughly 1,000-3,000 diverse labelled lines as an initial planning range; current handwriting output is unvalidated.

One AI mistake caught

Gemma3:4b and Gemma4:e2b returned repetitive Kannada/Devanagari instead of Khojki (37 s / 64 s; 2/2 wrong-script/format failures, no valid CER). Code-point inspection and the chart caught this. A render/Unicode verifier can reject wrong-script output but cannot identify every historical glyph; it was not used as an oracle for reinforcement learning.